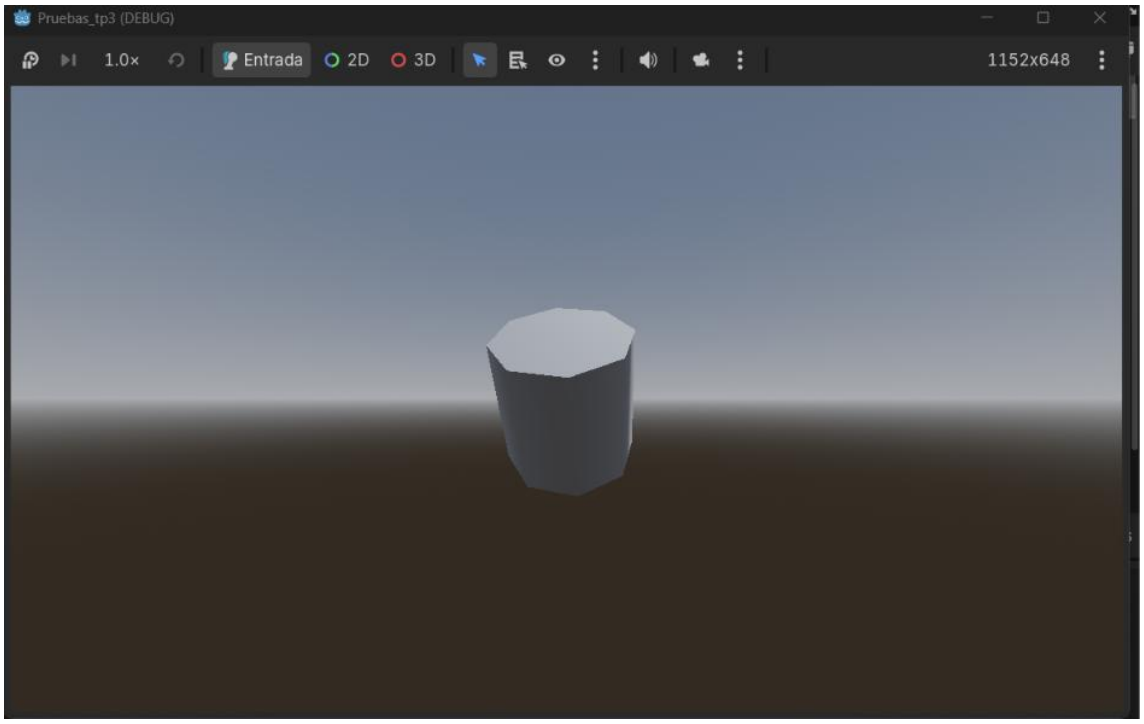
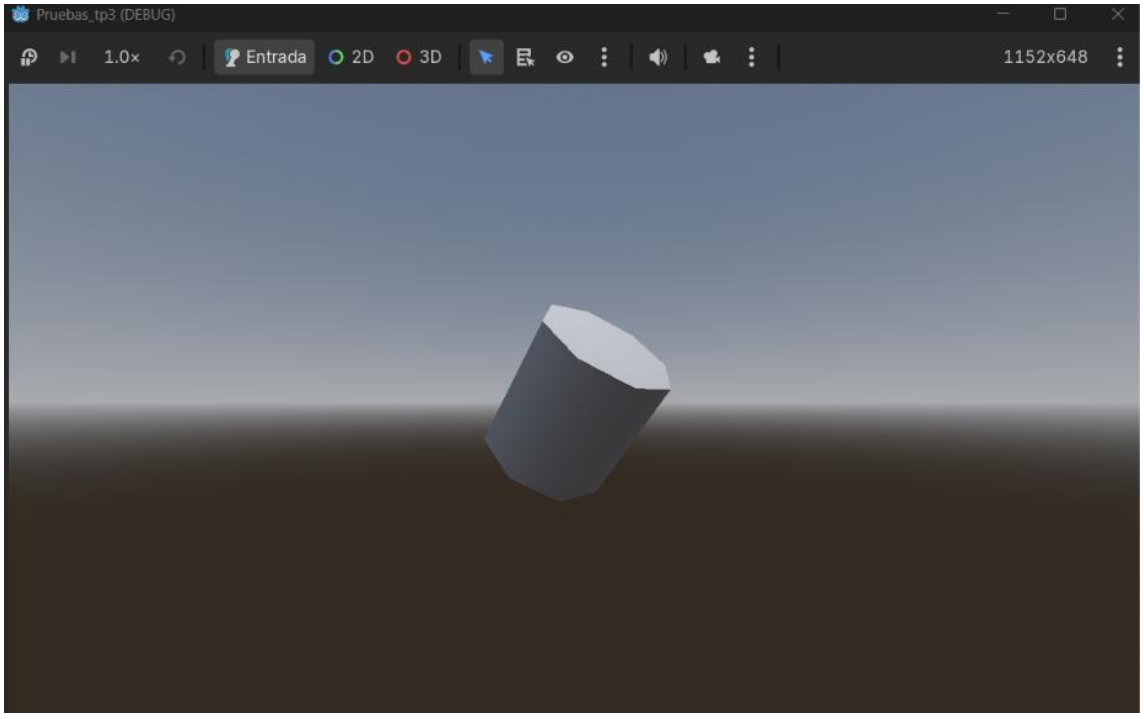
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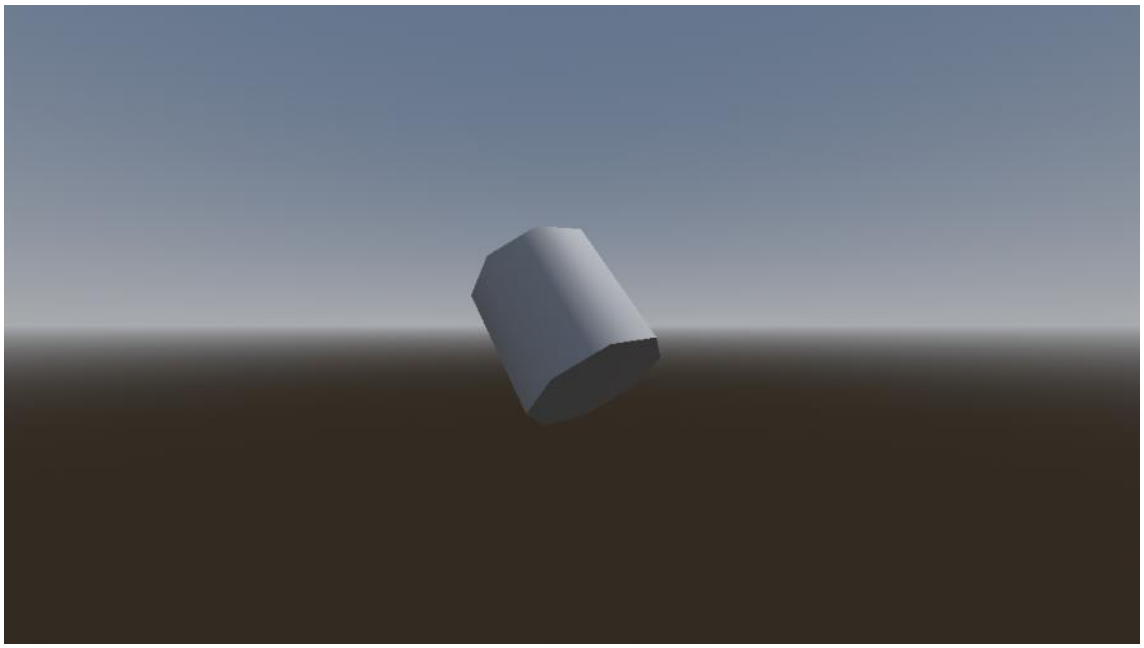
Trabajo práctico N°4

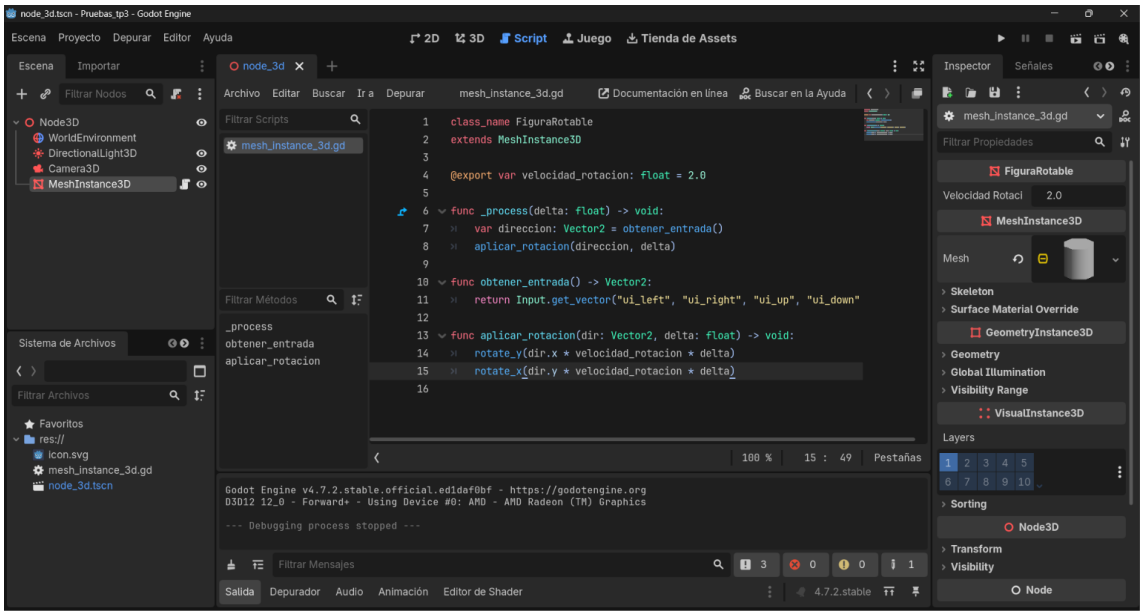
Ejercicio 2_a: Godot



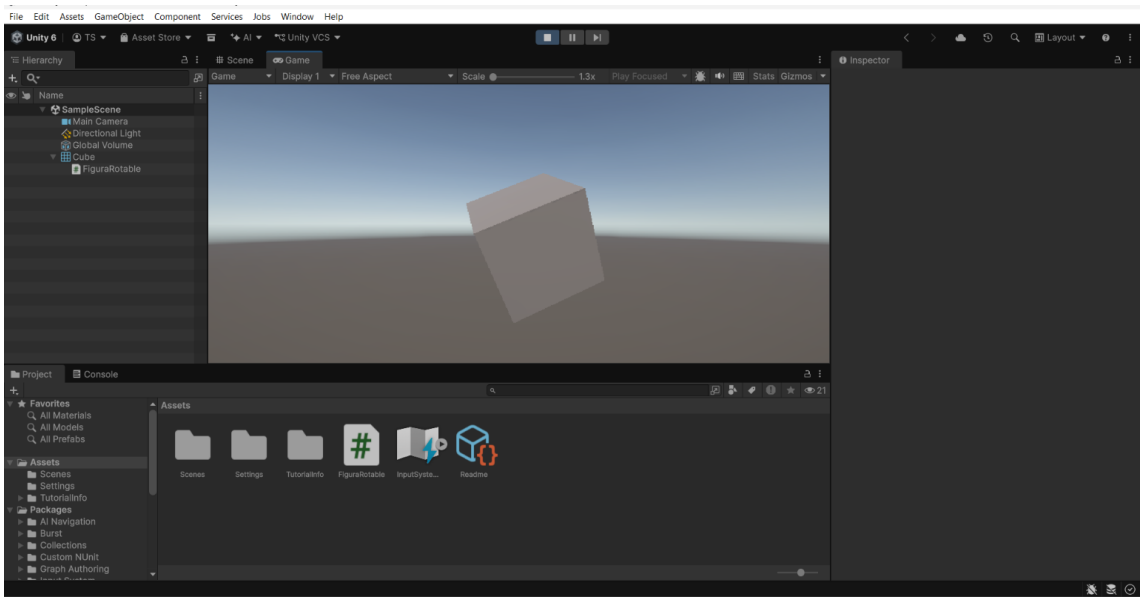


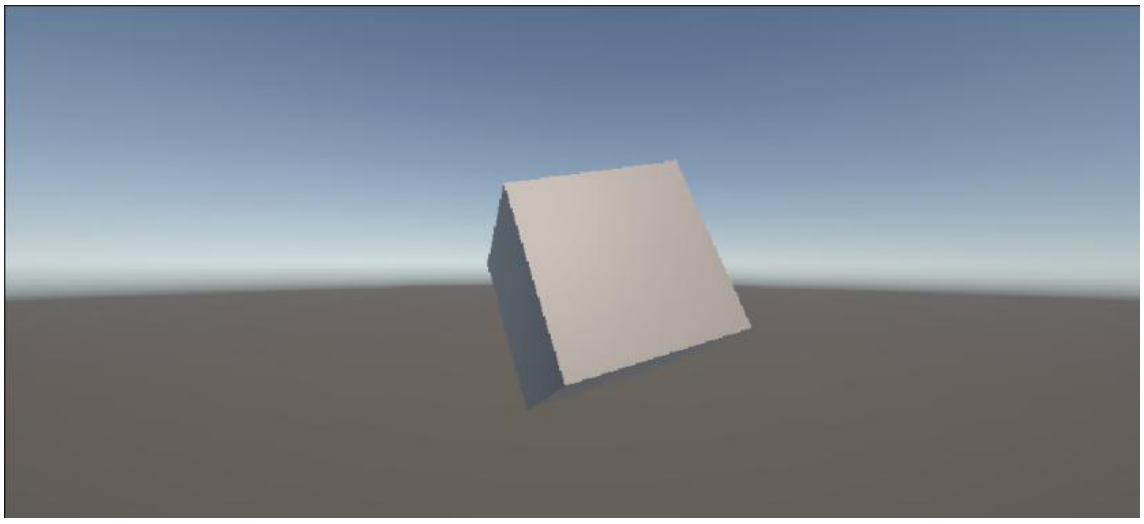
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Legajo: 14077

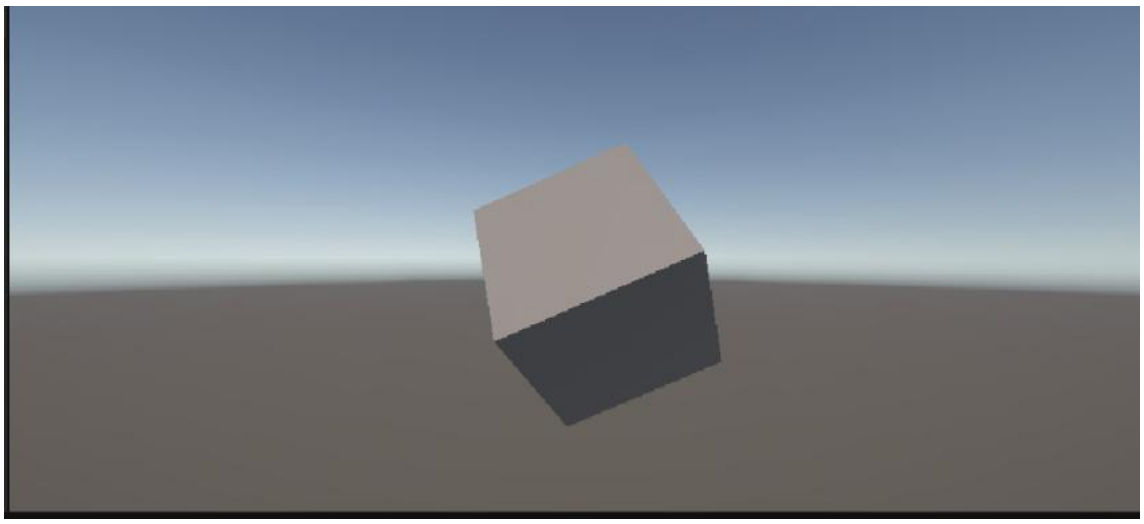




Ejercicio2_b: Unity


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FiguraRotable.cs

Assets > FiguraRotable.cs > ...

```
using UnityEngine;
using UnityEngine.InputSystem; // Necesario para Unity 6

// Clase POO encargada de leer el teclado y calcular el giro
3 referencias
public class ControladorEntrada
{
    2 referencias
    private float velocidad;

    1 referencia
    public ControladorEntrada(float vel)
    {
        velocidad = vel;
    }

    1 referencia
    public Vector2 ObtenerRotacion(float deltaTime)
    {
        float ejeHorizontal = 0f;
        float ejeVertical = 0f;

        var teclado = Keyboard.current;
        if (teclado != null)
        {
            if (teclado.leftArrowKey.isPressed)  ejeHorizontal = 1f;
            if (teclado.rightArrowKey.isPressed) ejeHorizontal = -1f;
            if (teclado.upArrowKey.isPressed)    ejeVertical = 1f;
            if (teclado.downArrowKey.isPressed)  ejeVertical = -1f;
        }

        return new Vector2(ejeVertical, ejeHorizontal) * velocidad * deltaTime;
    }
}

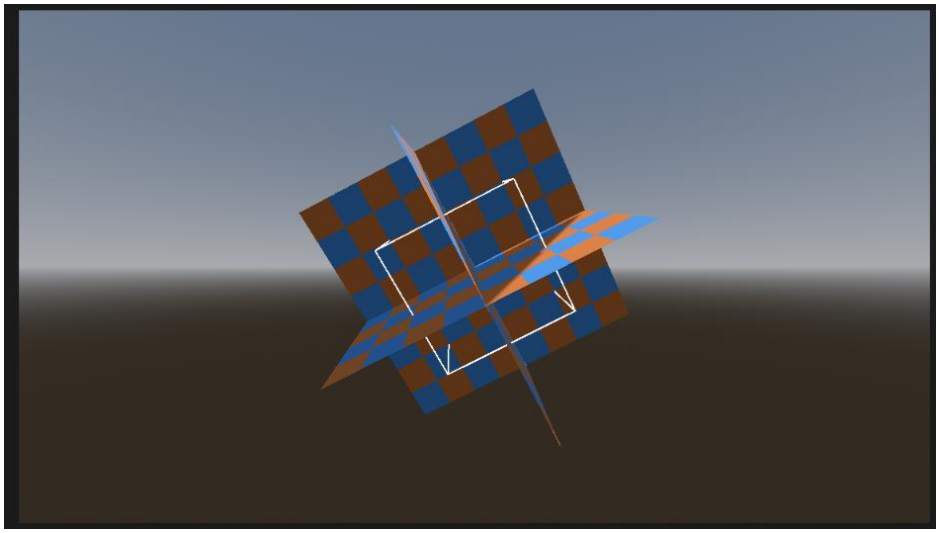
// Componente principal que se adjunta al Cubo
0 referencias | Script de Unity (2 referencias de recursos)
public class FiguraRotable : MonoBehaviour
{
    1 referencia | Campo serializado de Unity = 100
    public float velocidadRotacion = 100f;
    2 referencias
    private ControladorEntrada controlador;

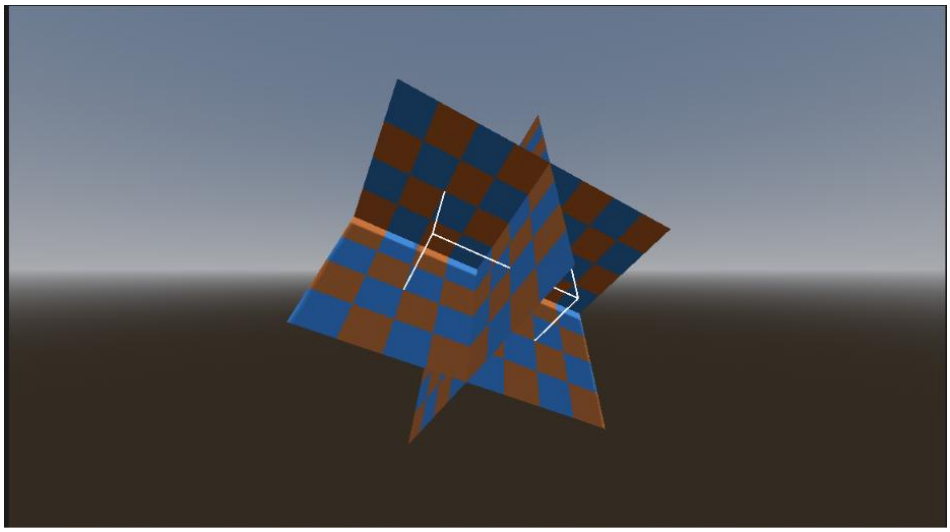
    0 referencias | Mensaje de Unity
    void Start()
    {
        controlador = new ControladorEntrada(velocidadRotacion);
    }

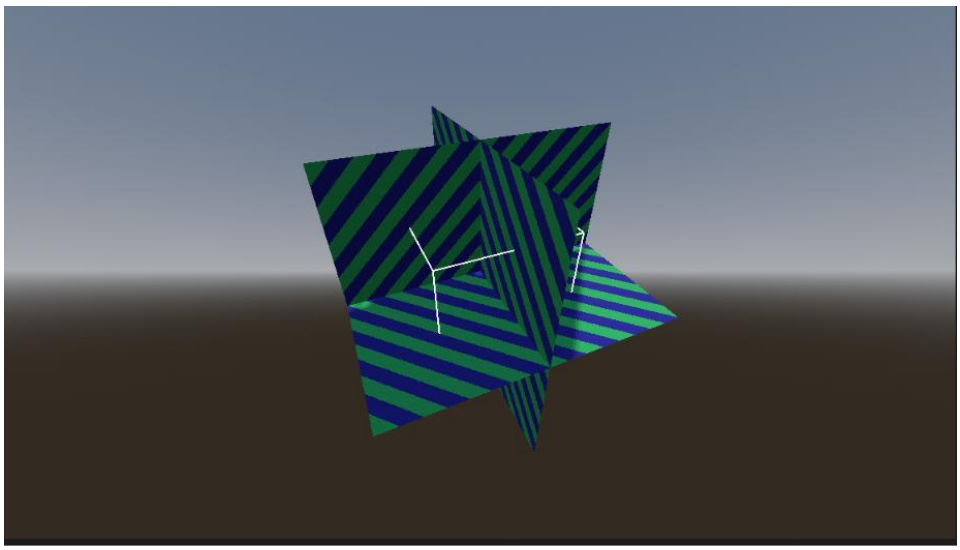
    0 referencias | Mensaje de Unity
    void Update()
    {
        Vector2 giro = controlador.ObtenerRotacion(Time.deltaTime);
        transform.Rotate(giro.x, giro.y, 0f, Space.World);
    }
}
```

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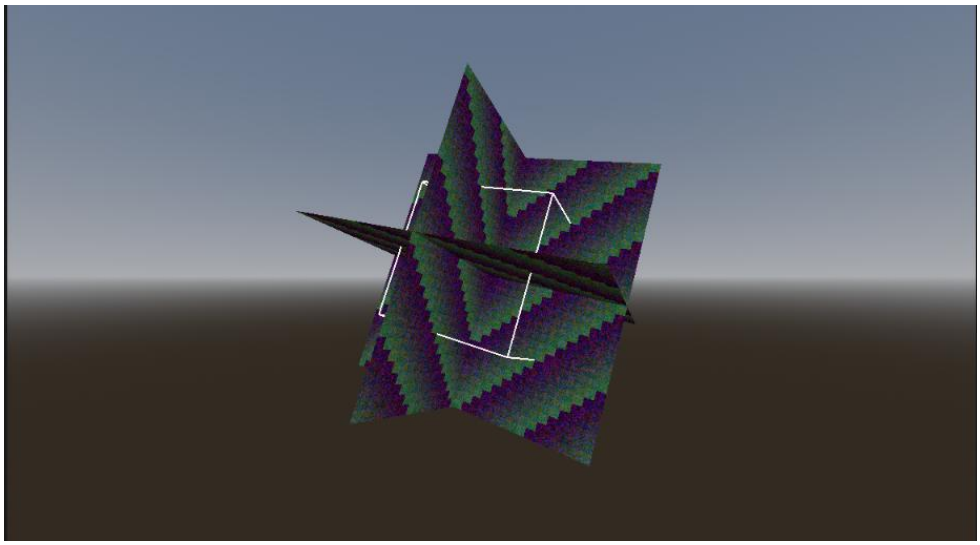
Ejercicio 3

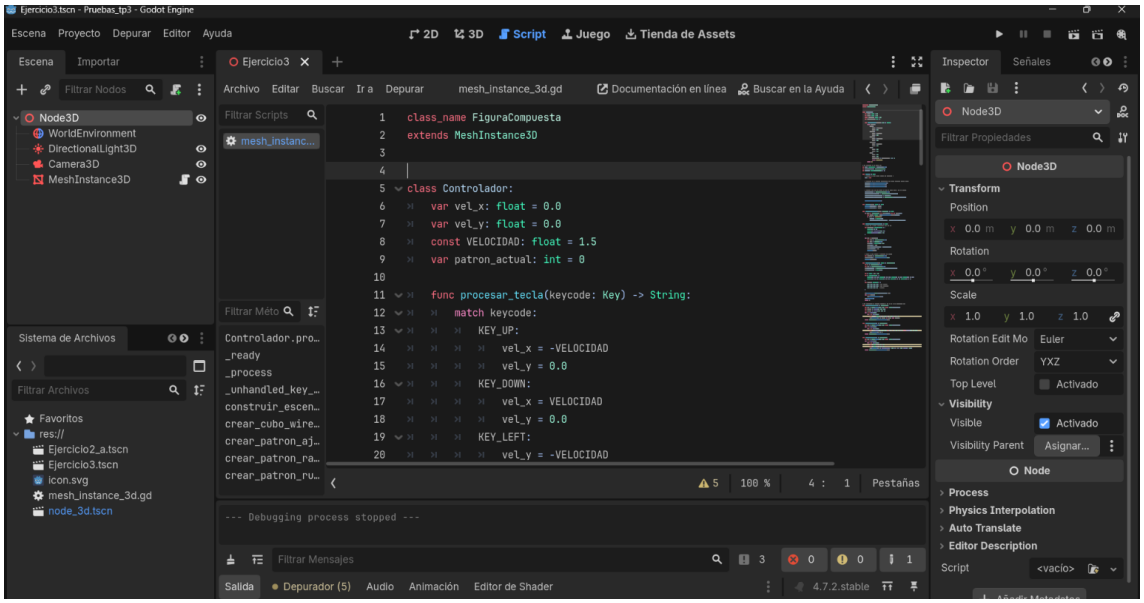






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```
21                    vel_x = 0.0
22                KEY_RIGHT:
23                    vel_y = VELOCIDAD
24                    vel_x = 0.0
25                KEY_D:
26                    vel_x = 0.0
27                    vel_y = 0.0
28                KEY_R:
29                    vel_x = 0.0
30                    vel_y = 0.0
31                    patron_actual = (patron_actual + 1) % 3
32                    return "reiniciar"
33            return ""
34
35    # --- Variables de la Figura ---
36    var ctrl: Controlador = Controlador.new()
37    var material_planos: StandardMaterial3D
38    var texturas: Array[ImageTexture] = []
39
```

```
40  func _ready() -> void:
41      # Quitamos el cubo blanco sólido del Ejercicio 2
42      mesh = null
43
44      texturas.append(crear_patron_ajedrez())
45      texturas.append(crear_patron_rayas())
46      texturas.append(crear_patron_ruido())
47
48      # Material para que los planos se vean de ambos lados y con tu textura
49      material_planos = StandardMaterial3D.new()
50      material_planos.cull_mode = BaseMaterial3D.CULL_DISABLED
51      material_planos.albedo_texture = texturas[0]
52      material_planos.texture_filter = BaseMaterial3D.TEXTURE_FILTER_NEAREST
53
54      construir_escena_processing()
55
56  func _process(delta: float) -> void:
57      rotate_x(ctrl.vel_x * delta)
58      rotate_y(ctrl.vel_y * delta)
59
```

```gdscript
func _unhandled_key_input(event: InputEvent) -> void:
    if event is InputEventKey and event.pressed and not event.echo:
        var accion: String = ctrl.procesar_tecla(event.keycode)
        if accion == "reiniciar":
            rotation = Vector3.ZERO
            material_planos.albedo_texture = texturas[ctrl.patron_actual]

func construir_escena_processing() -> void:
    # Crear los 3 planos ortogonales (XY, YZ, XZ)
    var rotaciones: Array[Vector3] = [
        Vector3(0, 0, 0),
        Vector3(0, deg_to_rad(90), 0),
        Vector3(deg_to_rad(90), 0, 0)
    ]

    for rot in rotaciones:
        var plano := MeshInstance3D.new()
        var quad := QuadMesh.new()
        quad.size = Vector2(2.2, 2.2)
        plano.mesh = quad
        plano.material_override = material_planos
        plano.rotation = rot
        add_child(plano)

    # Crear el cubo blanco en wireframe del centro
    var cubo_wire := MeshInstance3D.new()
    cubo_wire.mesh = crear_cubo_wireframe(1.1)
    add_child(cubo_wire)

func crear_cubo_wireframe(tam: float) -> ImmediateMesh:
    var im := ImmediateMesh.new()
    var mat := StandardMaterial3D.new()
    mat.shading_mode = BaseMaterial3D.SHADING_MODE_UNSHADED
    mat.albedo_color = Color.WHITE

    var m: float = tam / 2.0
    var v: Array[Vector3] = [
        Vector3(-m, -m, -m), Vector3(m, -m, -m), Vector3(m, m, -m), Vector3(-m, m, -m)
        Vector3(-m, -m, m), Vector3(m, -m, m), Vector3(m, m, m), Vector3(-m, m, m)
    ]
```

```
100     var aristas: Array[int] = [
101         0,1, 1,2, 2,3, 3,0, # Cara trasera
102         4,5, 5,6, 6,7, 7,4, # Cara delantera
103         0,4, 1,5, 2,6, 3,7  # Uniones
104     ]
105
106     im.surface_begin(Mesh.PRIMITIVE_LINES, mat)
107     for i in aristas:
108         im.surface_add_vertex(v[i])
109     im.surface_end()
110     return im
111
112 func crear_patron_ajedrez() -> ImageTexture:
113     var img := Image.create(256, 256, false, Image.FORMAT_RGBA8)
114     var celeste := Color8(80, 170, 255)
115     var naranja := Color8(255, 140, 60)
116     for x in range(256):
117         for y in range(256):
118             var es_par: bool = ((x / 32) + (y / 32)) % 2 == 0
119             img.set_pixel(x, y, celeste if es_par else naranja)
120     return ImageTexture.create_from_image(img)
```

```
121
122 func crear_patron_rayas() -> ImageTexture:
123     var img := Image.create(256, 256, false, Image.FORMAT_RGBA8)
124     for x in range(256):
125         for y in range(256):
126             var c := Color8(50, 220, 120) if ((x + y) / 16) % 2 == 0 else Color8(40, 40, 180)
127             img.set_pixel(x, y, c)
128     return ImageTexture.create_from_image(img)
129
130 func crear_patron_ruido() -> ImageTexture:
131     var img := Image.create(256, 256, false, Image.FORMAT_RGBA8)
132     for x in range(256):
133         for y in range(256):
134             var r: int = (x / 8) * 32
135             var g: int = (y / 8) * 32
136             img.set_pixel(x, y, Color8(randi() % 256, (r + g) % 256, randi() % 256))
137     return ImageTexture.create_from_image(img)
```